Codec-Aware, Quality-Gated Heart-Rate Estimation from Low-Quality Facial Video

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***Abstract*—We present a real-time heart-rate (HR) estimator for RGB facial video, designed around the deployment constraints of the 2027 SP Cup. A target-locked face front end samples six skin patches, the whole face and the background. The colour traces are cleaned of video-codec and illumination artefacts, turned into pulse spectra by the POS algorithm and by a 27k-parameter temporal CNN, fused by signal-to-noise ratio and tracked with a Viterbi decoder. A quality gate estimates, for every window, the probability that the error is within 5 bpm, and withholds readings below 0.8. We identify the main failure under the 10 fps / 500 kbps constraint: the B-frame cycle of default H.264 encoders puts a colour artefact at exactly 150 bpm. We remove it at both the sender and the receiver. The system was developed on 60 subjects of the public MCD-rPPG dataset and tested once on 20 subjects never used in development. Under normal conditions it reaches a mean absolute error (MAE) of 1.49 bpm ($r = 0.95$), against 5.84 bpm for POS. It stays at or below 2.4 bpm at QVGA resolution, at simulated 50 and 20 lux, with a bystander in frame, and when streamed at 10 fps and 500 kbps with our sender settings. In every condition, 94–98 % of the reported windows are within 5 bpm. Simulated surgical masks remain the main weakness (8.6 bpm); the system detects them and withholds most of those windows. Without retraining, on the UBFC-rPPG dataset (another lab, camera and lighting; 42 subjects) the MAE is 0.85 bpm under normal conditions and at most 0.95 bpm under simulated low light, low resolution, bystanders and masks; low-bandwidth streaming is harder there (3.2 bpm with our sender).**

***Index Terms*—remote photoplethysmography, heart rate, video compression, low light, signal quality, abstention**

I. Introduction

Remote photoplethysmography (rPPG) recovers the blood-volume pulse from small colour changes of facial skin [1], [2]. Under good conditions, simple projection methods such as CHROM [3] and POS [4] already track HR well. The FaceVitals challenge asks for more: HR from consumer cameras in low light, at low resolution, over 10 fps / 500 kbps links, with bystanders and masks, and with honest validation against a contact reference. Each of these conditions breaks a different assumption of the classical pipeline, and a method that only reports a pooled error hides where it fails.

This report describes our Round 1 system and how we evaluated it. Our contributions are:

- **A diagnosis of the low-bandwidth failure (C5).** With default H.264 settings, every fourth frame starts a new P B B B group; each position in the group is quantised differently, and at 10 fps this cycle is a 2.5 Hz (150 bpm) colour oscillation inside the HR band. Uneven frame decimation and the face-mesh refresh rate add further errors. We fix these at the sender (no B-frames, fixed decimation) and at the receiver (phase regression), without tuning on test data.
- **A hybrid estimator.** A 27k-parameter temporal CNN learns colour and region mixing from the traces and is fused with POS in the spectral domain. Out-of-fold, this cuts the error by 43–64 % in every unmasked condition.
- **Quality-gated reporting.** A small classifier predicts, per window, whether the error is within 5 bpm. Readings below a fixed threshold are flagged rather than reported, so that 94–98 % of reported windows are within 5 bpm in every tested condition.
- **A strict evaluation protocol.** Every number comes from subjects the scoring models never saw: subject-disjoint cross-validation for development, then two held-out sets, each scored only once, and a cross-dataset test on UBFC-rPPG with nothing retrained. We also report an ablation of every stage and results by skin tone, age and sex.

II. Related Work

Classical rPPG separates the pulse from specular and motion components by colour projections (CHROM [3], POS [4]) or by blind source separation [2]. Li *et al.* [5] used a background region to rectify illumination changes; our illumination step is a closed-form, per-window version of that idea. Deep models learn the projection from data, either from frames [6]–[8] or from pre-extracted traces; rPPG-Toolbox [9] collects many of them. McDuff *et al.* [10] showed that compression degrades rPPG, but did not analyse the effect of the B-frame cycle at low frame rates. Nowara *et al.* [11] found larger errors for darker skin, which motivates C2 and our stratified reporting. Our quality gate is a form of selective prediction [12]: trading coverage for reliability.

III. Method

Fig. 1 shows the pipeline. It runs both offline (recorded video) and online (webcam or stream), and the same code is used for both.

A. Face front end and regions of interest

YuNet [13] detects all faces at half resolution about 10 times per second (every third frame at 30 fps, every frame at 10 fps). A target tracker keeps one designated face. The target is chosen at start-up (the largest face, or the face nearest a selected point); after that, each detection round takes the

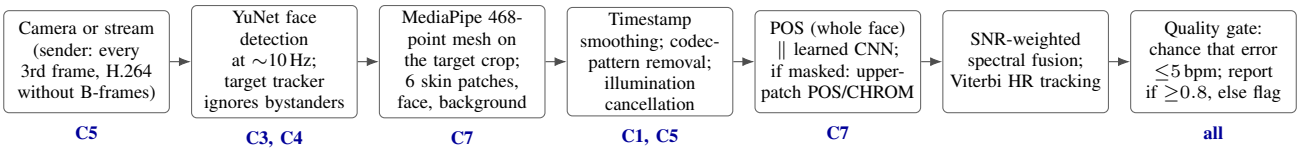


Fig. 1. Processing pipeline. Labels show the constraints each stage addresses.

box with the highest IoU (> 0.3) with the target's constant-velocity prediction. After more than five missed rounds it re-acquires only a face close to the last target position, so it never switches to a bystander. The MediaPipe face mesh [14] (468 landmarks) then runs on the target crop only, which keeps its cost independent of the number of people in view. From the mesh we sample the mean RGB of six skin patches (forehead, glabella, both malar cheeks and both lower cheeks), the whole face oval, and two background boxes beside the face at eye level. Saturated and black pixels are excluded.

B. Trace conditioning

Timestamps. Traces are placed on the real capture times. The webcam's logged stamps come in ∼16 ms timer steps, so we remove this jitter by averaging each stamp's deviation from a constant frame period over ∼1 s, which keeps genuine drift and dropped frames. The traces are then resampled to a uniform 30 Hz grid.

Codec-pattern removal. Let $\phi(n)$ be the number of frames since the last I or P frame (read from the stream). For each region and channel, the normalised trace $\tilde{c}(n) = c(n)/\mu - 1$ is high-passed with a moving average spanning whole cycles. It is then regressed on the zero-mean phase indicators $\mathbf{D} = [\mathbb{1}\{\phi(n) = k\}]_{k\geq 1}$, and the fitted offsets $\mathbf{D}\boldsymbol{\gamma}$ are removed from $\tilde{c}$. A stream without B-frames has a single phase and is left unchanged.

Illumination cancellation. Flicker, its aliases at low frame rates, and global light changes modulate skin and background alike, but only skin carries the pulse. With $\mathbf{B}$ the band-passed, normalised background RGB ($N \times 3$), each skin region becomes

$$\hat{\mathbf{c}}_r = \tilde{\mathbf{c}}_r - \mathbf{B}\boldsymbol{\beta}_r, \qquad \boldsymbol{\beta}_r = \arg\min_{\boldsymbol{\beta}} \left\| \mathcal{H}\tilde{\mathbf{c}}_r - \mathbf{B}\boldsymbol{\beta} \right\|^2, \tag{1}$$

where $\mathcal{H}$ is a third-order Butterworth band-pass over 0.66–3.5 Hz (40–210 bpm).

C. Pulse spectra

Classical branch. POS [4] (1.6 s overlap-add windows) is applied to the whole-face trace. GREEN and CHROM [3] are computed as well, but only as evidence for the quality gate. For each 10 s window (1 s hop), a Hann-tapered DFT gives the power $P(f)$ on a 1 bpm grid from 40 to 200 bpm.

Learned branch. A temporal CNN takes 27 input channels per 10 s window. For each of the eight regions it receives the mean-normalised, band-passed RGB; it also receives the whole-face POS, CHROM and GREEN signals. It has a 1×1 input layer (32 channels), five residual blocks of dilated convolutions (kernel 5, dilation 1–16, batch norm, ELU), dropout and a 1×1 output that emits a pulse waveform (27 073 parameters). It is trained with a spectral cross-entropy between the normalised power of its output, $\hat{P}$, and a Gaussian ($\sigma = 2.5$ bpm) centred on the contact-PPG heart rate:

$$\mathcal{L} = -\sum_f \mathcal{N}(f;\, \mathrm{HR}_{\mathrm{PPG}}, \sigma)\, \log \hat{P}(f). \tag{2}$$

This loss ignores the pulse transit delay between face and finger. Training uses the full 3-minute videos of the development subjects. Each video is time-stretched by factors 0.8–1.25 (with the label scaled accordingly) so that high heart rates are better covered. Further augmentation flips the sign, drops whole regions and adds noise. Training uses AdamW with a one-cycle schedule for 4000 steps and takes about 2 min on a laptop GPU.

Fusion and tracking. Each spectrum is normalised to unit power and weighted by its SNR [3]: the power within ±6 bpm of its own peak and ±12 bpm of the second harmonic, divided by the remaining power in the band. The fused spectrum is

$$\bar{P}_w(f) = \frac{\sum_k q_{k,w}\, P_{k,w}(f)}{\sum_k q_{k,w}}, \qquad q_{k,w} = 10^{\,\mathrm{SNR}_{k,w}/10}. \tag{3}$$

A Viterbi decoder then finds the most likely HR path. Its emissions are $\log \bar{P}_w$, and its transitions are Gaussian with $\sigma = 3$ bpm per second plus a small floor, so that genuine jumps remain possible. The online version runs the same recursion causally. We disabled the common harmonic boost: because the band ends at 210 bpm, rates above 105 bpm can never earn it while their half-rate sub-harmonics can, which biased the tracker toward half the true rate (Table II).

Mask gate (C7). A patch is marked occluded when its red-minus-blue chromaticity drops below half of the forehead's, or when its chromaticity is far from the forehead's. When both lower cheeks are occluded, the window switches from whole-face POS and the learned branch to SNR-fused POS and CHROM spectra of the patches that still look like skin.

D. Quality gate

For every window we compute seven features: the SNR, the share of spectral power within ±4 bpm of the estimate, the gap between the tracker output and the spectral peak, the gap between the POS and learned-branch peaks, the median power share over the surrounding 11 windows, and the gaps to the GREEN and CHROM peaks. A gradient-boosted classifier (depth 3, 200 iterations) maps them to $p_w = P(|e_w| \leq 5\,\mathrm{bpm})$. It is fitted on out-of-fold development windows; its AUC is 0.90 out-of-fold and 0.89 on the held-out subjects. A window is reported if $p_w \geq 0.8$; otherwise the HR

is still output but flagged as low quality. The threshold was fixed on the development subjects. Skin colour is deliberately not a feature.

IV. Experimental Setup

A. Data

We use the public MCD-rPPG dataset [15] (CC BY 4.0): 600 subjects, each recorded for 3 minutes at rest and again after exercise, with synchronised 100 Hz contact PPG and ECG. We use the frontal webcam (640×480, ∼30 fps). The reference HR is the spectral peak of the PPG in the same 10 s windows. For subjects that failed even under ideal conditions, we checked the reference by beat counting and against the ECG.

Subjects are ordered by ID. The first 60 (120 videos; 36 female; ages 18–78, median 20; reference HR 47–136 bpm) are the *development* set. All design choices were made on them with 5-fold subject-disjoint cross-validation, so every development number is out-of-fold. Two *held-out* sets were each scored once, with everything frozen. The first, subjects 41–60, was scored with models trained on subjects 1–40, before those subjects joined development. The second, subjects 61–80 (40 videos; 16 female; ages 18–50; HR 52–120 bpm), was scored with models trained on subjects 1–60, and gives our headline numbers. Evaluation uses the first 40 s of each video (30 windows per video).

For a cross-dataset test we use UBFC-rPPG DATASET_2 [18], whose authors share 42 of its 46 videos. It was recorded in another lab with a Logitech C920 webcam (640×480, uncompressed, ∼30 fps) about 1 m from the subject, indoors under varying sunlight and room light, while subjects played a time-pressured maths game to raise their heart rate. The reference is a CMS50E transmissive pulse oximeter, with one PPG sample per video frame. We use the whole videos (46–69 s, median 68 s), apply the same constraint simulations, and compute the reference HR the same way. The reference HR spans 53–135 bpm (mean 98 bpm, against 81–85 bpm on MCD).

B. Constraint simulations

All constraints are simulated on the same videos, which isolates each effect. *C1:* we undo the display gamma and scale the light by 50/300 or 20/300. We then draw Poisson shot noise (full well 3000 e^-) and Gaussian read noise (3 e^-), apply auto-gain back to the original brightness and quantise to 8 bits. This corresponds to the brief's "combined degradation" option. *C4:* we downsample to 320×240 (the QVGA stress case) with area interpolation. Note that the baseline is already VGA, below the 1280×720 the brief specifies for Task 1. *C5:* we keep every third frame (30→10 fps) and encode with x264 at 500 kbps (maxrate 500 kbps, buffer 1000 kb), both with default settings and with `zerolatency` tuning (our sender). *C3:* we place a second subject's video beside the target (1280×480); the target is designated once, at start-up. *C7:* we draw a light-blue textured surgical-mask polygon from the landmarks, from the mid nose bridge to the chin. It covers 49 % of the face oval and 96 % of its lower half (the brief requires ≥40 %).

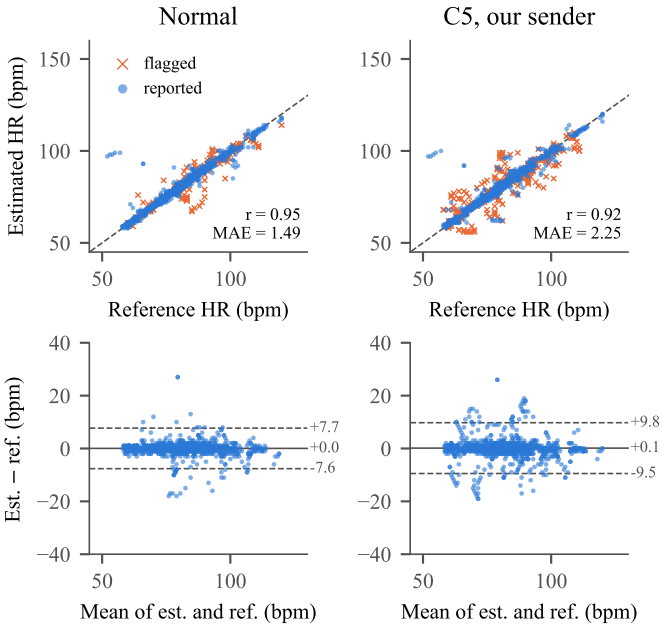


Fig. 2. Held-out subjects 61–80, proposed method. Top: estimated against reference HR per 10 s window (crosses: windows flagged by the quality gate). Bottom: Bland–Altman plots for all windows (bias and 95 % limits of agreement).

C. Metrics

Following the brief, we report the MAE, the RMSE, Pearson's r and the Bland–Altman bias with 95 % limits of agreement [16]. All are computed over pooled 10 s windows. We also report the share of windows within 5 bpm and, with the quality gate, the coverage (the share of windows reported) and the error on reported windows.

V. Results

A. Accuracy on held-out subjects

Table I gives the headline results. Under normal conditions the proposed system has an MAE of 1.49 bpm ($r = 0.95$, Bland–Altman bias +0.04 bpm, limits −7.6 to +7.7 bpm), against 5.84 bpm for POS and 3.00 bpm for our classical branch alone. The system degrades gracefully: the MAE stays at or below 2.4 bpm at QVGA, at simulated 50 and 20 lux, with a bystander in frame, and over a 10 fps / 500 kbps link with our sender settings. Fig. 2 shows the agreement plots. On the earlier held-out set (subjects 41–60, model trained on subjects 1–40) the normal-condition MAE was 2.65 bpm (classical 3.14, POS 5.87). That set has more fast, post-exercise heart rates, so results should be compared between methods on the same set, not across the two sets.

With the quality gate, 73–85 % of windows are reported in every unmasked condition except default x264 (65 %). The reported windows have an MAE of 1.2–2.2 bpm, and 94–98 % of them are within 5 bpm. The gate is most useful under masks: it reports 37 % of windows, and 97 % of those are within 5 bpm.

TABLE I

HELD-OUT RESULTS: SUBJECTS 61–80 (20 SUBJECTS, 40 VIDEOS, 1200 WINDOWS PER CONDITION), SCORED ONCE WITH MODELS TRAINED ON SUBJECTS 1–60. MAE, RMSE AND BLAND–ALTMAN (BA) VALUES IN BPM. "REPORTED" COLUMNS APPLY THE QUALITY GATE ($p \geq 0.8$).

	MAE, all windows			Proposed, all windows				Proposed, reported windows		
Condition	POS	Classical	Proposed	RMSE	r	BA bias [95 % LoA]	$\leq$5 bpm (%)	Reported (%)	MAE	$\leq$5 bpm (%)
Normal (VGA, 30 fps)	5.84	3.00	**1.49**	3.91	0.95	0.04 [-7.6, 7.7]	95	85	1.21	98
C4: QVGA 320×240	6.46	3.20	**1.73**	4.55	0.93	0.18 [-8.7, 9.1]	93	82	1.22	97
C1: 50 lux (sim.)	6.66	4.69	**2.14**	5.49	0.91	0.66 [-10.0, 11.3]	92	83	1.29	98
C1: 20 lux (sim.)	7.49	3.95	**2.38**	5.44	0.92	1.25 [-9.1, 11.6]	88	76	1.17	96
C3: bystander in frame	6.08	3.35	**1.44**	3.81	0.95	0.12 [-7.3, 7.6]	96	84	1.27	97
C5: 10 fps, 500 kbps, our sender	9.99	4.78	**2.25**	4.90	0.92	0.14 [-9.5, 9.8]	89	73	1.61	95
C5: 10 fps, 500 kbps, default x264	12.73	6.34	**4.04**	9.00	0.77	2.16 [-15.0, 19.3]	83	65	2.19	94
C7: surgical mask (sim.)	10.48	**8.49**	8.56	15.07	0.41	-6.06 [-33.1, 21.0]	66	37	1.25	97

TABLE II

ABLATION ON THE DEVELOPMENT SUBJECTS (1–60, 120 VIDEOS), OUT-OF-FOLD: EACH STAGE IS ADDED IN TURN. POOLED MAE IN BPM. THE LAST TWO ROWS EACH CHANGE ONE DESIGN CHOICE OF THE CLASSICAL SYSTEM.

Stage	Normal	C4 QVGA	C1 50 lux	C1 20 lux	C3 bystander	C5 our sender	C5 default x264	C7 mask
POS, spectral peak	7.15	7.22	7.76	8.96	7.25	11.12	22.18	13.66
+ Viterbi tracking	5.35	4.83	5.18	5.80	5.06	7.38	19.67	9.74
+ illumination cancellation	5.00	4.88	4.09	6.08	5.07	6.63	16.64	**8.70**
+ codec-pattern removal	5.00	4.88	4.09	6.08	5.07	6.63	12.33	**8.70**
+ mask gate (classical final)	4.99	4.87	4.09	6.08	5.06	6.63	12.33	9.99
+ learned branch (proposed)	**2.21**	**2.66**	**2.33**	**3.14**	**2.37**	**3.62**	**4.49**	9.99
Classical final, harmonic boost on	5.78	6.74	5.67	6.76	5.68	7.68	14.05	11.20
Classical final, no codec removal	4.99	4.87	4.09	6.08	5.06	6.63	16.65	9.99

B. Where the gains come from

Table II adds one stage at a time on the development set. Viterbi tracking removes isolated wrong peaks, cutting the normal-condition MAE from 7.15 to 5.35 bpm. Illumination cancellation helps most at 50 lux and over compressed links, but costs a little at 20 lux. Codec-pattern removal acts only on streams that contain B-frames, where it lowers the error from 16.6 to 12.3 bpm. The learned branch gives the largest single gain: it cuts the error by 43–64 % in every unmasked condition (normal 5.0→2.2 bpm; default x264 12.3→4.5 bpm). Turning the tracker's harmonic boost back on makes every condition worse.

The mask gate did *not* lower the pooled error on simulated masks (9.99 against 8.70 bpm for whole-face POS with tracking and illumination cancellation). A likely reason is that the simulated mask is a static texture with no pulse: averaging it into the face dilutes the pulse without adding a competing signal, which favours whole-face POS. A real mask moves with breathing and speech. The gate still detects the mask in 100 % of masked windows, with 0.9 % false alarms in other conditions (development set). That signal lets the quality gate withhold unreliable windows instead of failing silently, as C7 requires. Better compensation is planned for Round 2.

C. Low-bandwidth operation (C5)

The first run at 10 fps / 500 kbps failed in a characteristic way: many videos locked onto 150 bpm regardless of the true rate. Separating frame rate from bitrate on the development subjects traced the failure to the four causes in Table III. Fig. 3 shows the main one. Within ±3 bpm of 150 bpm, the mean spectrum of default-x264 video holds 9.6 % of the power, against 2.9 % at 30 fps. Codec-pattern removal lowers this to 4.1 %, and our sender to 3.8 %. The remaining bump is common to all conditions and is not specific to compression. The fixes were frozen before the held-out sets were scored. If the organisers' C5 videos are encoded with default settings, only the receiver-side fix applies, giving 4.04 bpm on held-out subjects; with our sender settings the error is 2.25 bpm.

TABLE III

WHAT BROKE 10 FPS / 500 KBPS, AND THE FIXES (FOUND ON DEVELOPMENT SUBJECTS).

Cause	Symptom	Fix
Default x264 repeats P B B B; each position quantises the face colour differently	false peak at $10/4 = 2.5$ Hz (150 bpm)	sender: no B-frames (`zerolatency`); receiver: phase regression
Picking the frame nearest each 0.1 s tick from jittery stamps gives an uneven 2/3/4-frame pattern; with 100 Hz light this aliases into the band	the same 150 bpm lock, even without B-frames	keep exactly every 3rd frame; smooth the stamps
Face mesh refreshed every 3rd *frame*: only 3.3 Hz at 10 fps	regions drift during motion	refresh at ∼10 Hz in real time
Tracker harmonic boost	fast HR read at half rate	boost disabled

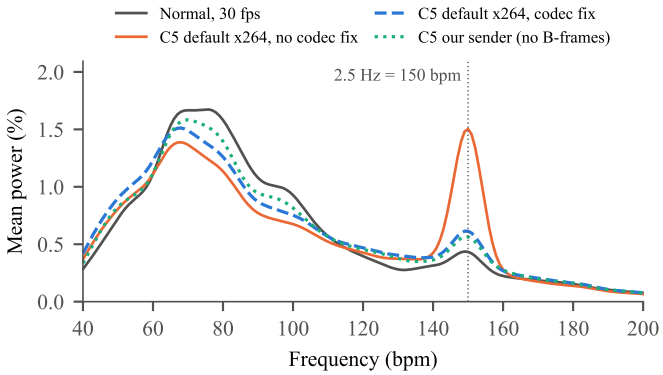


Fig. 3. Mean normalised pulse spectrum over the 120 development videos. The default x264 stream at 10 fps shows a peak at 150 bpm that does not depend on the subject's heart rate; codec-pattern removal and our sender remove most of it.

TABLE IV
NORMAL CONDITION BY SUBGROUP: 80 SUBJECTS (1–60 OUT-OF-FOLD, 61–80 HELD-OUT). MAE IN BPM; Δ IS THE DIFFERENCE FROM THE POOLED MAE.

Group	Subj.	MAE	Δ	≤5 (%)	Reported (%)	MAE rep.
Measured ITA, relative (uncalibrated)						
lightest third	27	2.62	+0.60	91	83	1.40
middle third	26	1.61	-0.42	94	84	1.03
darkest third	27	1.84	-0.19	94	89	1.47
Age (years)						
18–29	73	1.91	-0.11	93	85	1.28
30–59	3	7.03	+5.01	74	78	3.65
≥60	4	0.38	-1.65	100	100	0.38
Sex						
F	52	1.77	-0.25	94	84	1.14
M	28	2.50	+0.47	91	88	1.60
All	80	2.03	–	93	85	1.31

D. Bystanders (C3)

In all 40 held-out videos with a bystander, both faces were detected in every frame, and the target was found in 99.9 % of frames. In every frame where both runs found the face, the target's colour trace matched the single-person run within 5 %, so the tracker never switched to the bystander. The error is unchanged (1.44 against 1.49 bpm).

E. Results by skin tone, age and sex (C2, C6)

Table IV follows the brief's ITA-based stratification [17]. Measured ITA did not predict error: the darkest third by ITA had an MAE of 1.84 bpm, against 2.03 bpm overall. However, the absolute ITA values cannot be trusted. Because the webcam exposes faces dimly ($L^* \approx 40$), the measured ITA ranges from $-59°$ to $+8°$, which would put every subject in Fitzpatrick V–VI. We therefore report relative thirds only. Proper C2 claims need colour-calibrated video and a dataset with genuinely dark skin; we plan both for Round 2.

The four subjects aged 63–78 had an MAE of 0.38 bpm, and all their windows were reported. That supports C6, but the group is far too small to establish it. The 30–59 group's error comes from a single subject (age 54, MAE 20.1 bpm). In that subject the pulse is barely present in any region: about 10 % of the spectral power lies at the true rate, against 30–40 % normally. The gate withholds 65 % of those windows.

F. Cross-dataset test on UBFC-rPPG

To check that the system works beyond the data it was built on, we ran it unchanged on UBFC-rPPG (Table V). Under normal conditions the MAE is 0.85 bpm ($r = 0.99$, Bland–Altman bias $+0.10$ bpm, limits -3.9 to $+4.1$ bpm), and only one of the 42 subjects has an MAE above 5 bpm (5.3 bpm). Every simulated condition except low bandwidth stays below 1 bpm. That includes masks (0.85 bpm): the mask was drawn and detected in every window of all 42 videos, but these faces keep a strong pulse on the forehead and upper cheeks. UBFC is easier than MCD for every method (POS: 0.91 bpm), since its faces are well lit and the video is uncompressed. The learned branch adds little here, but it does not hurt.

Low bandwidth is the exception. With our sender the MAE is 3.24 bpm (POS 9.15 bpm). With default x264 settings the classical branch fails (32.7 bpm). In 25 of the 28 subjects whose heart rate averages 90 bpm or more, it locks onto 43–85 bpm, while the four subjects below 75 bpm average 2.6 bpm. We have not yet confirmed the mechanism. One candidate is that at 500 kbps the B-frames carry little new colour information, which lowers the effective sampling rate of the colour signal. The learned branch recovers part of the loss (fused 10.0 bpm), and the quality gate withholds 45 % of those windows. This reinforces the main C5 lesson: at 10 fps, the sender's encoder settings decide whether HR survives.

[TO DO: Results on a dataset with darker skin tones (Annexure I), in a table like Table V.]

G. Live operation and run time

The live application (Fig. 4) shows each stage as it runs, so a user can see why a reading is reported or withheld. It uses the causal tracker and applies the quality gate to the newest window. We replayed a held-out subject for 92 s, switching to simulated 20 lux and then to a mask part-way through. The live error was 1.3 bpm, and 81 % of readings were reported.

On one Apple M3 Pro process, the front end (detection, mesh and region sampling) takes 5.5 ms per frame at 30 fps and 13.1 ms per frame at 10 fps, where the mesh runs on every frame. Each HR update takes 11.6 ms, including the learned branch, and the online tracker step is negligible. With one update per second, the pipeline uses about 18 % of one core at 30 fps and 14 % at 10 fps, so it runs in real time on a laptop. We expect it to fit a Raspberry Pi 5, which has four cores, but have not measured it yet. **[TO DO: measured Raspberry Pi 5 timings]**

VI. COMPLIANCE AND DELIVERABLES

Table VI summarises the constraints. Five (C1, C3, C4, C5, C7) are addressed and quantified on two datasets, and two (C2, C6) are reported in a preliminary form. `process_round1.py` produces the Round 1 package in `/Round1_Baseline/`:

TABLE V

CROSS-DATASET RESULTS: UBFC-RPPG DATASET_2 (42 SUBJECTS, WHOLE VIDEOS, 2390 WINDOWS PER CONDITION), RUN WITH THE SYSTEM FROZEN ON MCD-RPPG SUBJECTS 1–60; NOTHING WAS RETRAINED OR RE-TUNED. COLUMNS AS IN TABLE I.

	MAE, all windows			Proposed, all windows				Proposed, reported windows		
Condition	POS	Classical	Proposed	RMSE	r	BA bias [95 % LoA]	$\leq$5 bpm (%)	Reported (%)	MAE	$\leq$5 bpm (%)
Normal (VGA, 30 fps)	0.91	**0.83**	0.85	2.05	0.99	0.10 [-3.9, 4.1]	98	98	0.70	99
C4: QVGA 320×240	1.03	0.88	**0.88**	2.08	0.99	0.05 [-4.0, 4.1]	98	97	0.71	99
C1: 50 lux (sim.)	1.09	**0.88**	0.90	2.04	0.99	0.03 [-4.0, 4.0]	98	97	0.74	99
C1: 20 lux (sim.)	1.51	1.07	**0.95**	2.19	0.99	0.04 [-4.3, 4.3]	98	96	0.78	99
C3: bystander in frame	0.96	0.87	**0.85**	2.03	0.99	0.08 [-3.9, 4.1]	98	98	0.73	99
C5: 10 fps, 500 kbps, our sender	9.15	4.11	**3.24**	8.64	0.89	-2.21 [-18.6, 14.2]	89	79	1.38	97
C5: 10 fps, 500 kbps, default x264	29.77	32.67	**10.04**	18.57	0.52	-7.59 [-40.8, 25.6]	65	55	4.20	88
C7: surgical mask (sim.)	2.03	**0.85**	**0.85**	2.21	0.99	-0.06 [-4.4, 4.3]	98	92	0.60	100

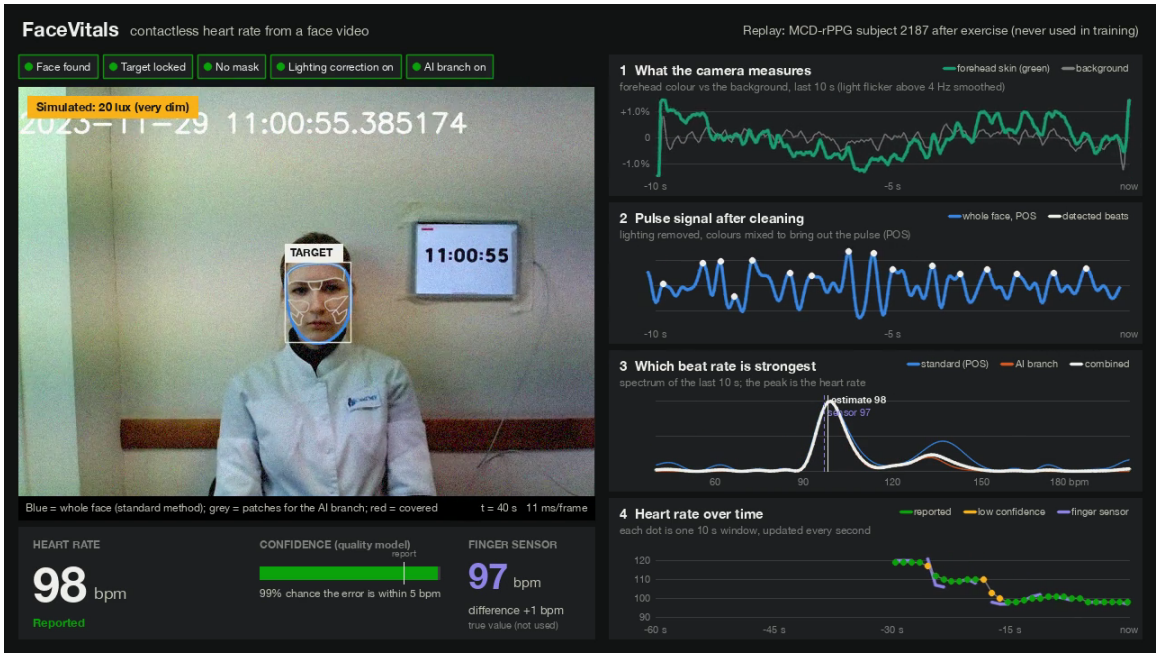


Fig. 4. Live dashboard on a held-out MCD-rPPG subject [15] (CC BY 4.0) under simulated 20 lux. Left: the target face (whole-face region in blue, skin patches in grey) and the heart-rate card with the quality gate's confidence and, for dataset replays, the finger-sensor reference. Right, in pipeline order: forehead and background colour, the cleaned pulse with detected beats, the POS, learned-branch and combined spectra, and the HR history (green: reported; orange: withheld).

`Round1_Baseline_None.mat` (variables `facial_video`, `roi_data`, `reference_signal`, `processed_signal`, `metrics` and `params`), `target_video.mp4`, `reference_signal.csv` and `processed_output.csv`. For each window it also writes p_w and whether the window was reported. **[TO DO: store raw RGB frames in `facial_video` (currently 72×72 face crops) and fill lux, distance and Fitzpatrick type in `params` for any recording that has them]**.

Data and ethics. Round 1 uses only public data: MCD-rPPG (CC BY 4.0) and UBFC-rPPG, which its authors share for research use and ask to be cited. Following the UBFC readme, no UBFC face images appear in this report. We collected no new human data, so no IRB approval was needed for this round. **[TO DO: if the team records its own videos for Round 2, attach IRB approval and consent forms]**. All code was written from scratch, without third-party rPPG code, to avoid licence conflicts.

VII. LIMITATIONS AND ROUND 2 PLAN

- Low light, masks and bystanders are simulated, and both datasets are indoor recordings of mostly young adults. The next step is real recordings: real low-light video, a darker-skin dataset from Annexure I and, if possible, the organisers' hospital data.
- Default H.264 at 10 fps and 500 kbps fails for fast heart rates (UBFC). We will test the B-frame hypothesis and, if it holds, recommend no-B-frame encoding as part of the system.
- C2 needs colour-calibrated skin-tone measurement (for example, a grey card) and per-type tolerances set before testing.

TABLE VI
ROUND 1 CONSTRAINT COMPLIANCE. "SIM." MEANS SIMULATED WITH THE PROCEDURE IN SECTION IV-B. EVIDENCE GIVES THE MAE IN BPM ON MCD-RPPG HELD-OUT SUBJECTS AND, AFTER THE SLASH, ON UBFC-RPPG.

	Specification	How it is addressed	Evidence: MAE, MCD held-out / UBFC	Status
C1	$\leq$50 lux at the face	illumination cancellation against background; learned branch; quality gate	50 lux: 2.14 / 0.90; 20 lux: 2.38 / 0.95 (POS at 20 lux: 7.49 / 1.51)	sim.
C2	Fitzpatrick I–VI, per-type error	skin colour excluded from the gate; ITA stratification	no trend with measured ITA; absolute ITA unreliable (Section V-E)	partial
C3	track one target among several faces	detector + IoU tracker; mesh on the target only	no identity switches in 40 videos; 1.44 / 0.85	sim.
C4	$\leq 640\times480$; QVGA stress case	everything runs at VGA; QVGA tested	VGA baseline 1.49 / 0.85; QVGA 1.73 / 0.88	sim.
C5	$\leq$10 fps, $\leq$500 kbps; offline mode	sender without B-frames; codec-pattern removal; on-device pipeline	our sender 2.25 / 3.24; default x264 4.04 / 10.04	sim.
C6	ages 60–70+, reported separately	no special handling	4 subjects aged 63–78: MAE 0.38 (dev., out-of-fold)	partial
C7	$\geq$40 % of lower face covered	mask detection; switch to upper patches; quality gate	8.56 / 0.85; detected in every masked window; MCD: 37 % reported at 1.25	sim.

- Simulated masks: the learned branch is not yet trained on masked faces. We will train it with mask augmentation and test against real masks.
- Very weak pulses, such as the 54-year-old subject above, are not recovered by any branch; the system abstains on them instead.
- Talking and smiling. Our benchmarks score the first 40 s of each video, while subjects sit still. Later in one held-out video, where the subject smiles and moves, the live error rose to 13 bpm. Two thirds of those readings were withheld, but one wrong reading (79 against 123 bpm) passed the gate. Motion features for the gate and motion-robust regions are planned.
- Round 2 (SpO_2): the same front end and regions give the per-channel AC/DC ratios needed for ratio-of-ratios estimation. The quality gate will be extended to SpO_2.

VIII. CONCLUSION

A classical rPPG core, cleaned of codec and illumination artefacts, fused with a very small learned branch and protected by a quality gate, estimates heart rate from low-quality facial video. On held-out subjects, its error is 1.4–2.4 bpm under normal conditions and under simulated low light, low resolution, bystanders and low-bandwidth streaming. Run unchanged on a second dataset (UBFC-rPPG), it reaches 0.85 bpm. The biggest practical lesson concerns C5: at 10 fps the encoder's frame structure, not only its bitrate, decides whether the pulse survives, and a single encoder setting removes most of the damage. Code, trained models and every result table are reproducible from the accompanying scripts.

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